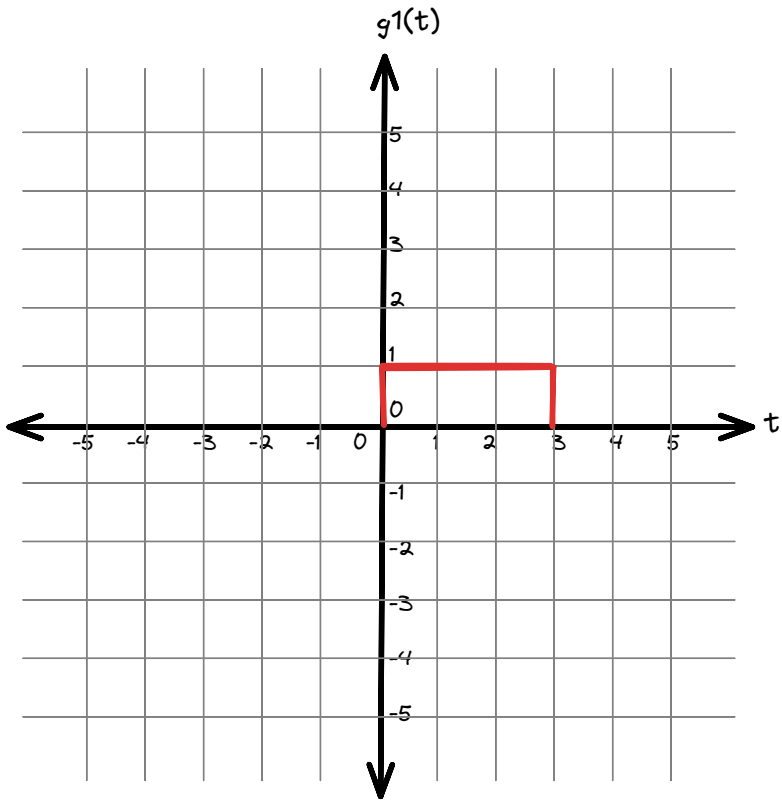
Final Exam

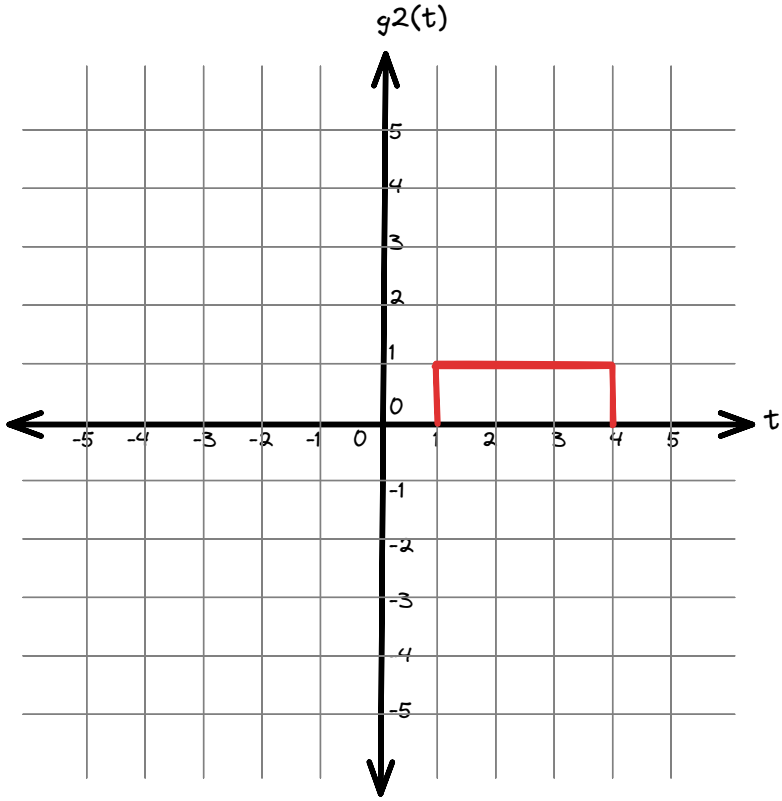
Question 1

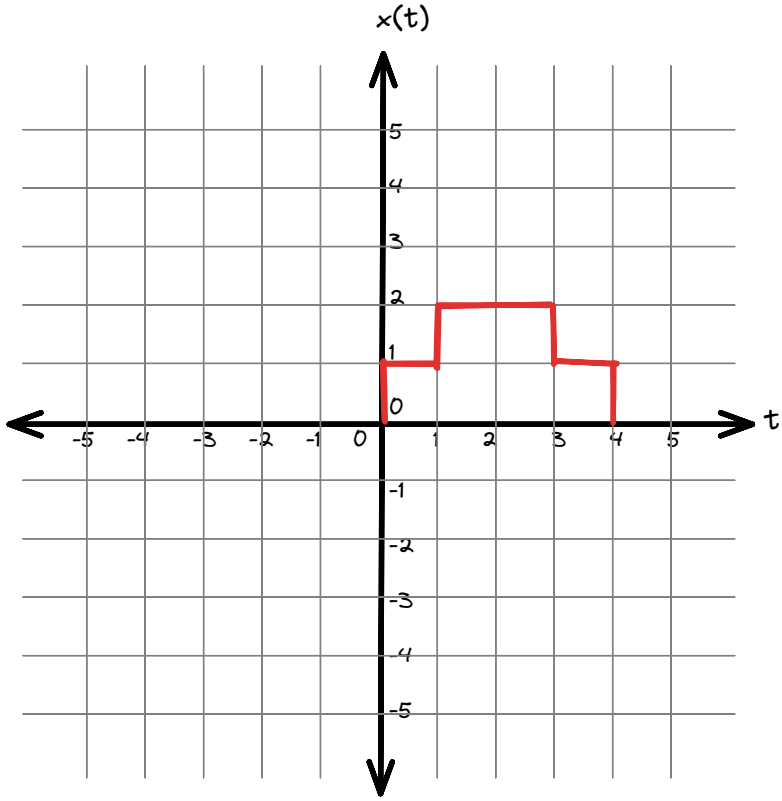
(a)

With the definitions $g_1(t) = g\left(\frac{t}{3} - \frac{3}{2}\right)$ and $g_2(t) = g\left(\frac{t}{3} - \frac{5}{2}\right)$, the signal can be constructed as

$$x(t) = g_1(t) + g_2(t) = g\left(\frac{t}{3} - \frac{3}{2}\right) + g\left(\frac{t}{3} - \frac{5}{2}\right)$$




g2(t)
t
5
4
3
2
1
0
-1
-2
-3
-4
-5
-5 -4 -3 -2 -1 0 1 2 3 4 5


x(t)
t
5
4
3
2
1
0
-1
-2
-3
-4
-5
-5 -4 -3 -2 -1 0 1 2 3 4 5

(b)

(i)

Due to the $u(t)$ factor, it will not be able to repeat the sinusoidal pattern below $t = 0$. It is therefore not periodic.

(ii)

Whether it is periodic depends on if there exists and integer m that makes the term inside the cosine be equal to an integer multiple of 2π which will be equal to the fundamental period (denoted N in this case). This can be found as follows

$$\begin{aligned} \frac{5\pi N}{2} &= 2\pi m, \quad m \in \mathbb{Z} \\ N &= \frac{4\pi}{5\pi} m \\ N &= \frac{4}{5} m \\ N &= 4, \quad m = 5 \end{aligned}$$

Since both m and N can make the equation be true while staying as integers the signal is periodic with a period of $N = 4$.

(iii)

Even though it is a continuous time signal, the signal does not depend on the time t, which makes it a constant signal, meaning that it always repeats itself. It must therefore be a periodic signal, but with no tangible fundamental period or a fundamental period of 0.

(c)

Memoryless
Since the system depends on a past value at time $t-1$, it is not memoryless.

Causal
Since it still depends on the same past value, it is not causal.

Time-invariant
Shifting the input leads to

$$x\left((t-1)-t_0\right)=x(t-t_0-1)$$

Inserting this into the output signal gives

$$y(t)=tx(t-t_0-1)$$

However shifting the output signal leads to the following expression

$$\begin{aligned} y\left(t-t_0\right) &= (t-t_0)x((t-t_0)-1) \\ &= tx(t-t_0-1)-t_0x(t-t_0-1) \\ &\neq tx(t-t_0-1) \end{aligned}$$

Since they do not lead to the same expressions, the system is not time-invariant.

Linear
A linear combination of the input would be

$$x_3(t)=ax_1\left(t\right)+bx_2\left(t\right)$$

Using this in the output would result in

$$\begin{aligned} y_3(t) &= tx_3\left(t\right) \\ &= t(ax_1(t-1)+bx_2(t-1)) \\ &= tax_1(t-1)+tbx_2(t-1) \end{aligned}$$

A linear combination of the output would lead to the following expression

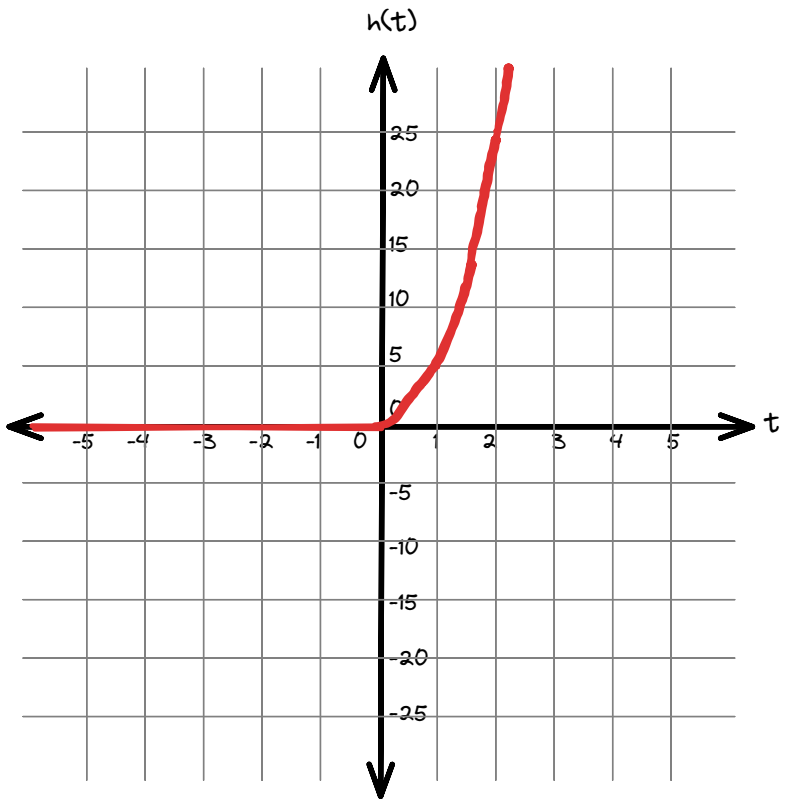
$$\begin{aligned} y_3(t) &= ay_1(t)+by_2(t) \\ &= tax_1\left(\frac{t}{2}\right)+tbx_2\left(\frac{t}{2}\right) \end{aligned}$$

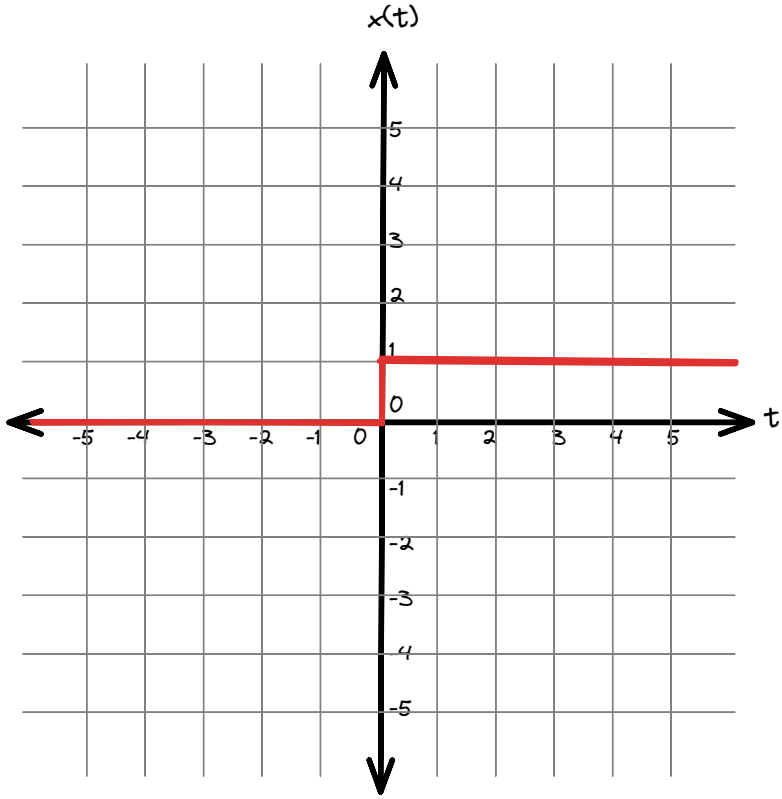
Since they do lead to the same expression the system is linear.

Question 2

(a)

(i)





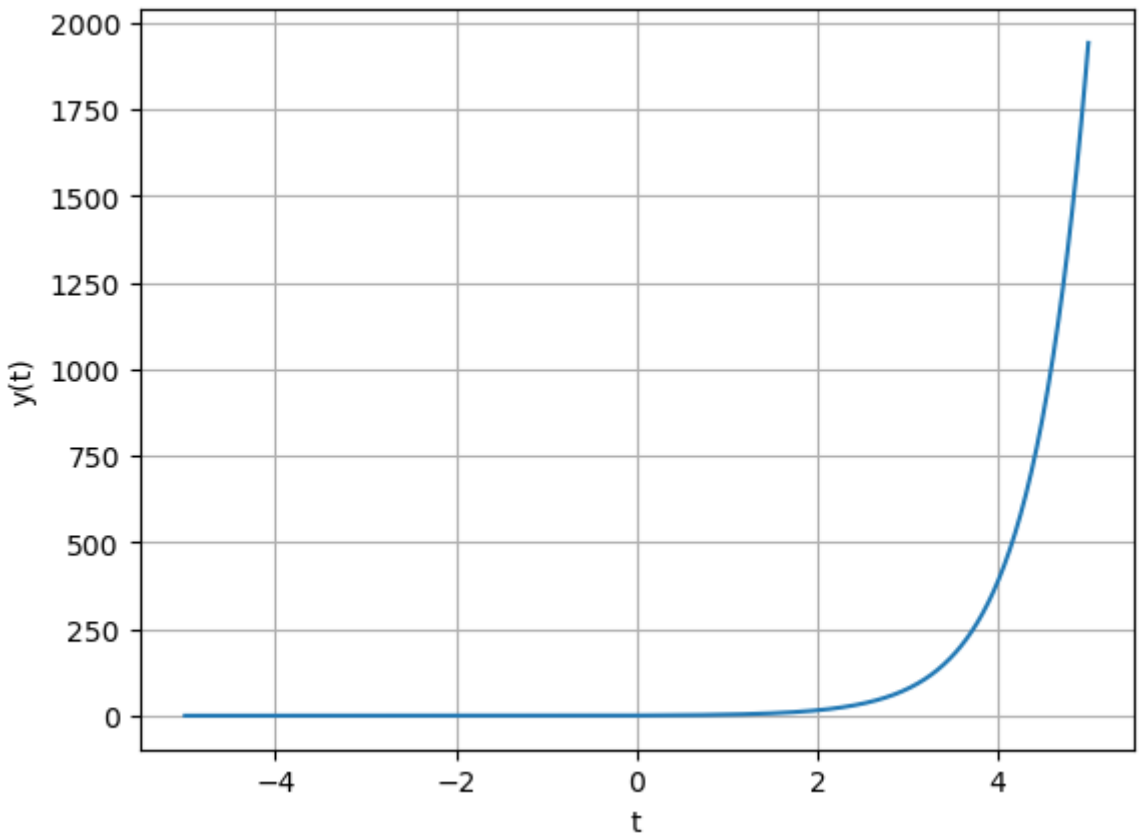
(ii)

$$
\begin{aligned}
y(t) &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau \\
&= \int_{-\infty}^{\infty} u(\tau)5^{t-\tau}u(t-\tau)\,d\tau \\
&= \int_{0}^{t} 5^{t-\tau}\,d\tau \\
&= -\frac{1}{\log(5)}\left[5^{t-\tau}\right]_0^t \\
&= -\frac{1}{\log(5)}\left[5^{t-t} - 5^{t-0}\right] \\
&= \frac{5^t - 1}{\log(5)}
\end{aligned}
$$

(iii)

The plot was made in python using the following code

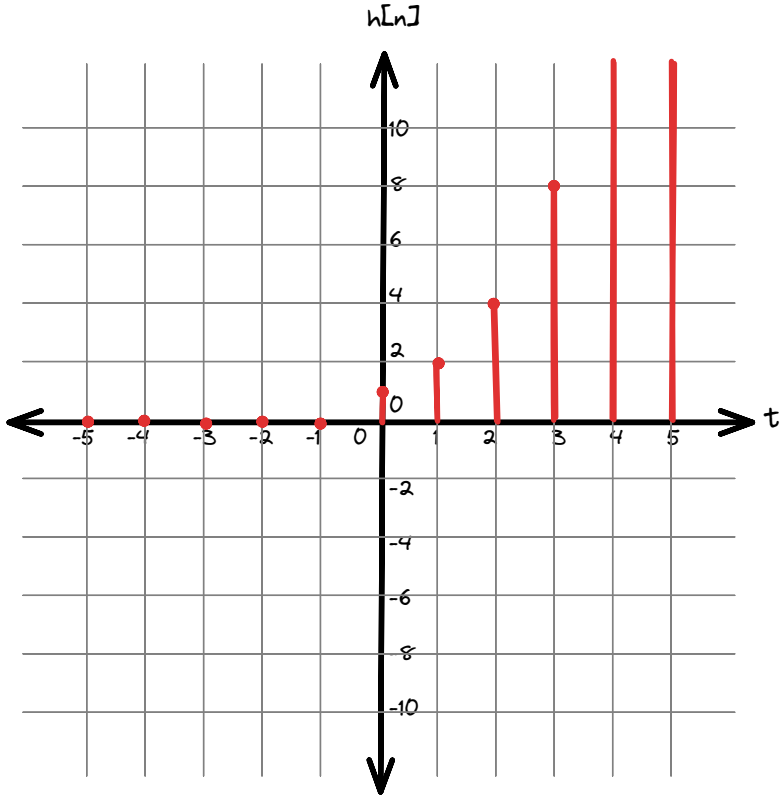
```
t = np.linspace(-5, 5, 1000)
x = (5**t - 1) / np.log(5) # Assuming natural logarithm
plt.plot(t, x)
plt.grid()
```

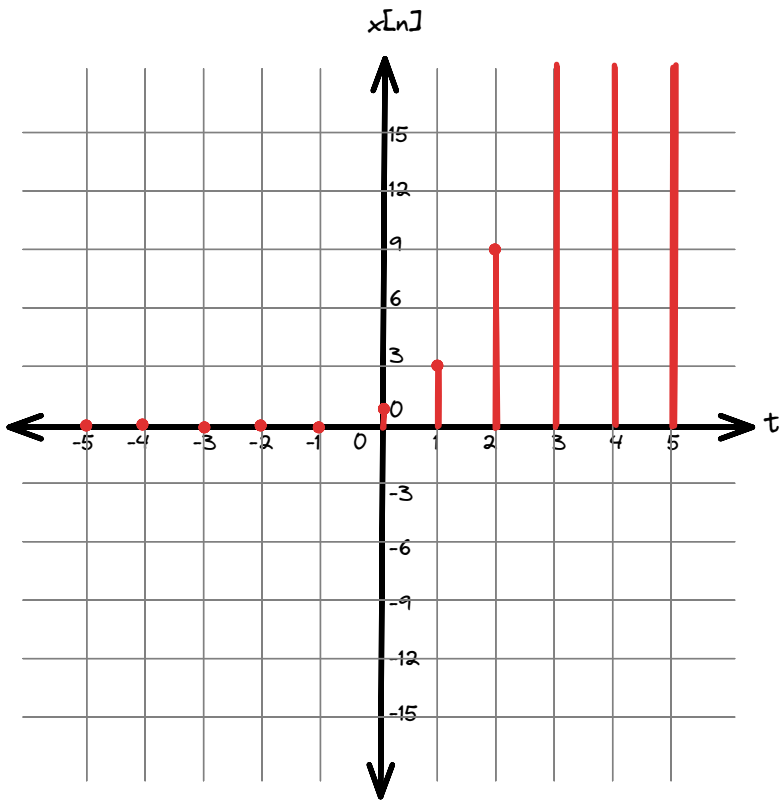


(b)

(i)

The top of some of the datapoints of each function are not drawn since they go too far up (these do not have a dot at the top of the line)


h[n]
10
8
6
4
2
0
-2
-4
-6
-8
-10
-5
-4
-3
-2
-1
0
1
2
3
4
5
t


x[n]
15
12
9
6
3
0
-3
-6
-9
-12
-15
-5
-4
-3
-2
-1
0
1
2
3
4
5
t

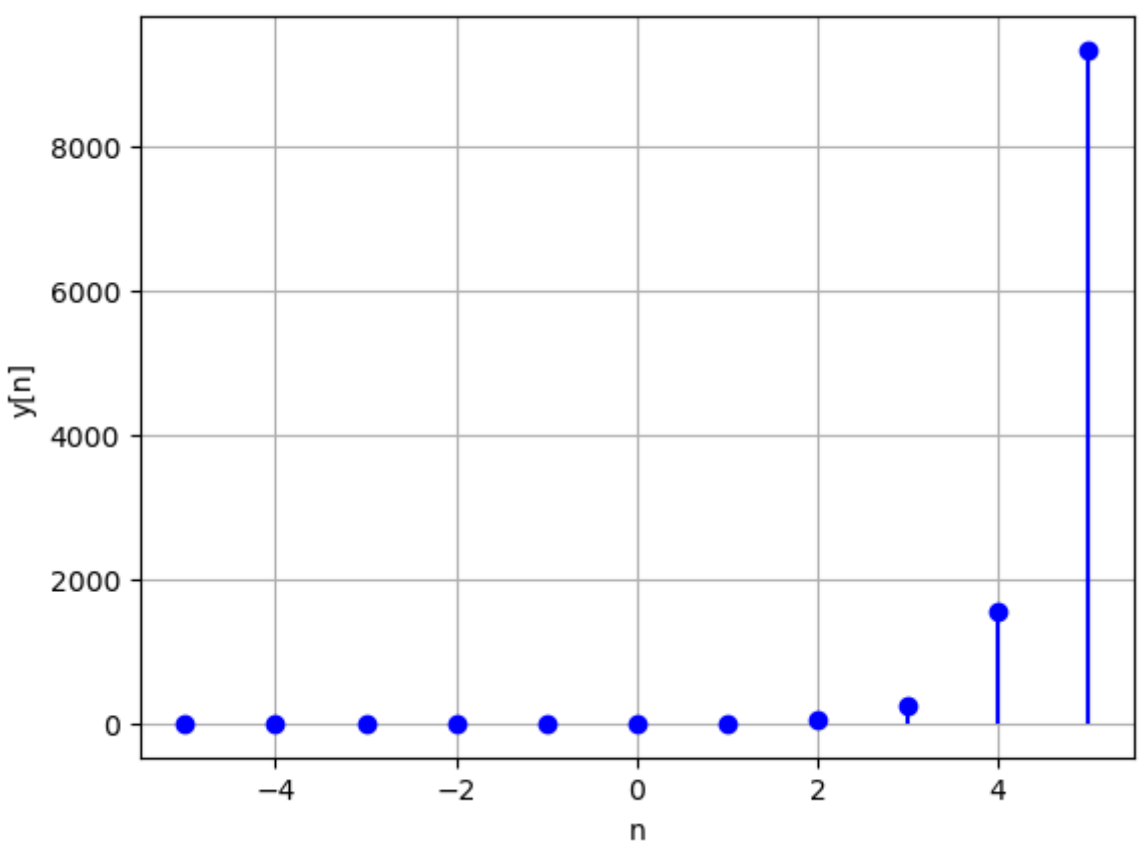
(ii)

$$
\begin{aligned}
y[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\
&= \sum_{k=-\infty}^{\infty} 3^k u[k] 2^k u[n-k] \\
&= \sum_{k=0}^{n} 3^k 2^k \\
&= \frac{1}{5}(6^{n+1} - 1)
\end{aligned}
$$

(iii)

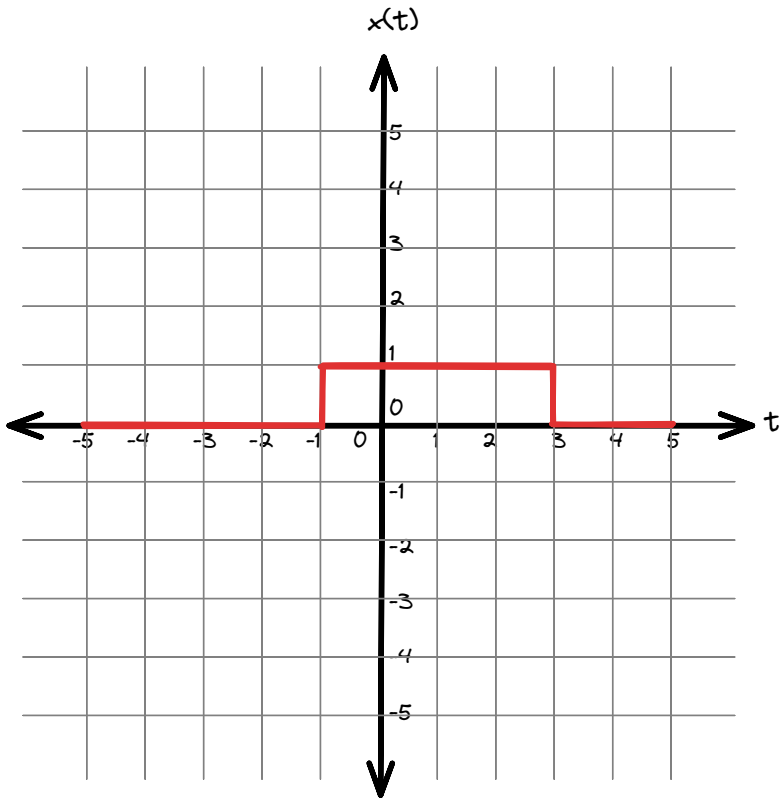
The plot was made in python using the following code:

```
n = np.linspace(-5, 5, 11)
y = np.piecewise(n, [n < 0, n >= n], [0, 1/5 * (6**(n + 1) - 1)])
plt.plot(n, y, "bo")
for i in n:
    plt.vlines(i, 0, int(y[int(i+5)]), colors="b")
plt.xlabel("n")
plt.ylabel("y[n]")
plt.grid()
```



Question 3

(i)



(ii)

$$
\begin{aligned}
X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt \\
&= \int_{-1}^{3} e^{-j\omega t}\,dt \\
&= -\frac{1}{j\omega}[e^{-j\omega t}]_{-1}^{3} \\
&= -\frac{1}{j\omega}[e^{-3j\omega} - e^{j\omega}] \\
&= -\frac{e^{-3j\omega} - e^{j\omega}}{j\omega} \\
&= \frac{e^{j\omega} - e^{-3j\omega}}{j\omega} \\
&= \frac{\cos(\omega) + j\sin(\omega) - \cos(3\omega) + j\sin(3\omega)}{j\omega}
\end{aligned}
$$

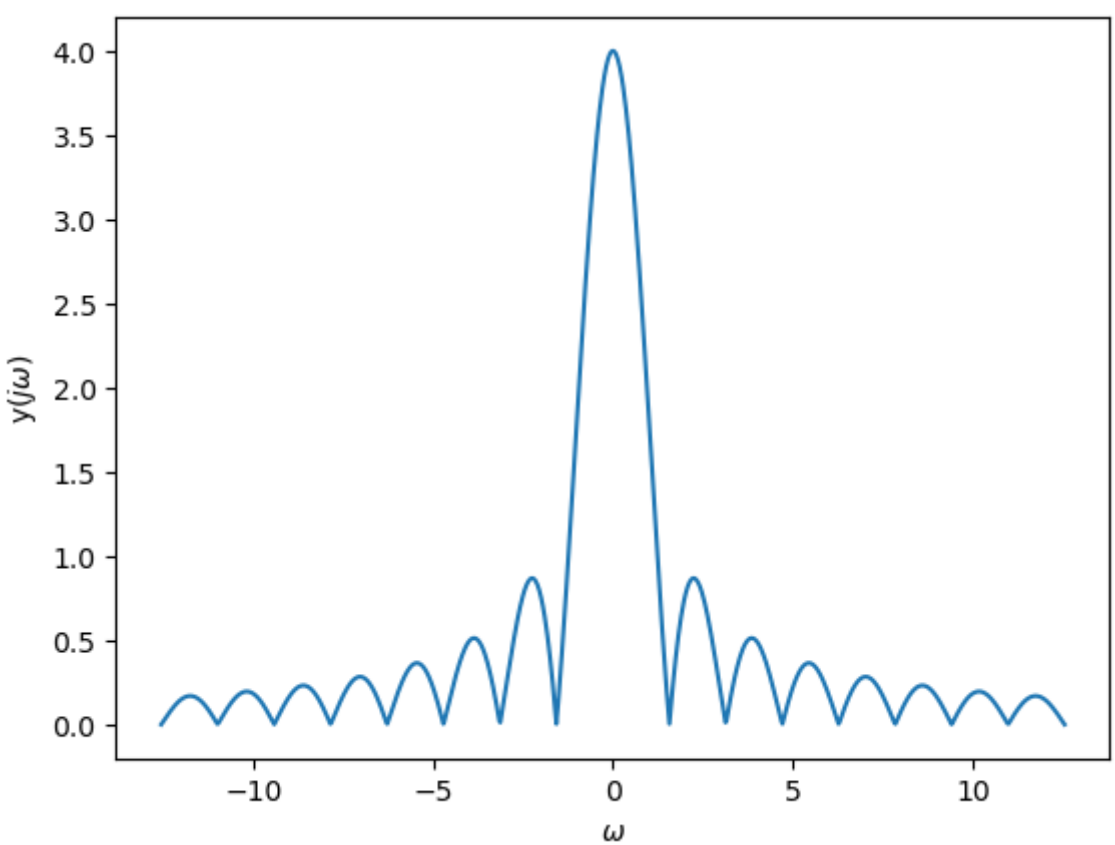
(iii)

$$
\begin{aligned}
|X(j\omega)| &= \left| \frac{\cos(\omega) + j\sin(\omega) - \cos(3\omega) + j\sin(3\omega)}{j\omega} \right| \\
&= \frac{|\cos(\omega) - \cos(3\omega) + j\sin(\omega) + j\sin(3\omega)|}{|j\omega|} \\
&= \frac{\sqrt{(\cos(\omega) - \cos(3\omega))^2 + (\sin(\omega) + \sin(3\omega))^2}}{\sqrt{\omega^2}} \\
&= \frac{\sqrt{\cos^2(\omega) + \cos^2(3\omega) - 2\cos(\omega)\cos(3\omega) + \sin^2(\omega) + \sin^2(3\omega) + 2\sin(\omega)\sin(3\omega)}}{|\omega|}
\end{aligned}
$$

(iv)

The sketch is done in python using the following code:

```
omega = np.linspace(-4*np.pi, 4*np.pi, 1000)
X = np.sqrt((np.cos(omega) - np.cos(3*omega))**2 + (np.sin(omega) +
np.sin(3*omega))**2) / np.abs(omega)
plt.xlabel("$\omega$")
plt.ylabel("y($j\omega$)")
plt.plot(omega, X)
```



Question 4

(a)

(i)

The frequency response is found using equation 4.76.

$$H(j\omega) = \frac{j\omega + 1}{2j\omega + 5}$$

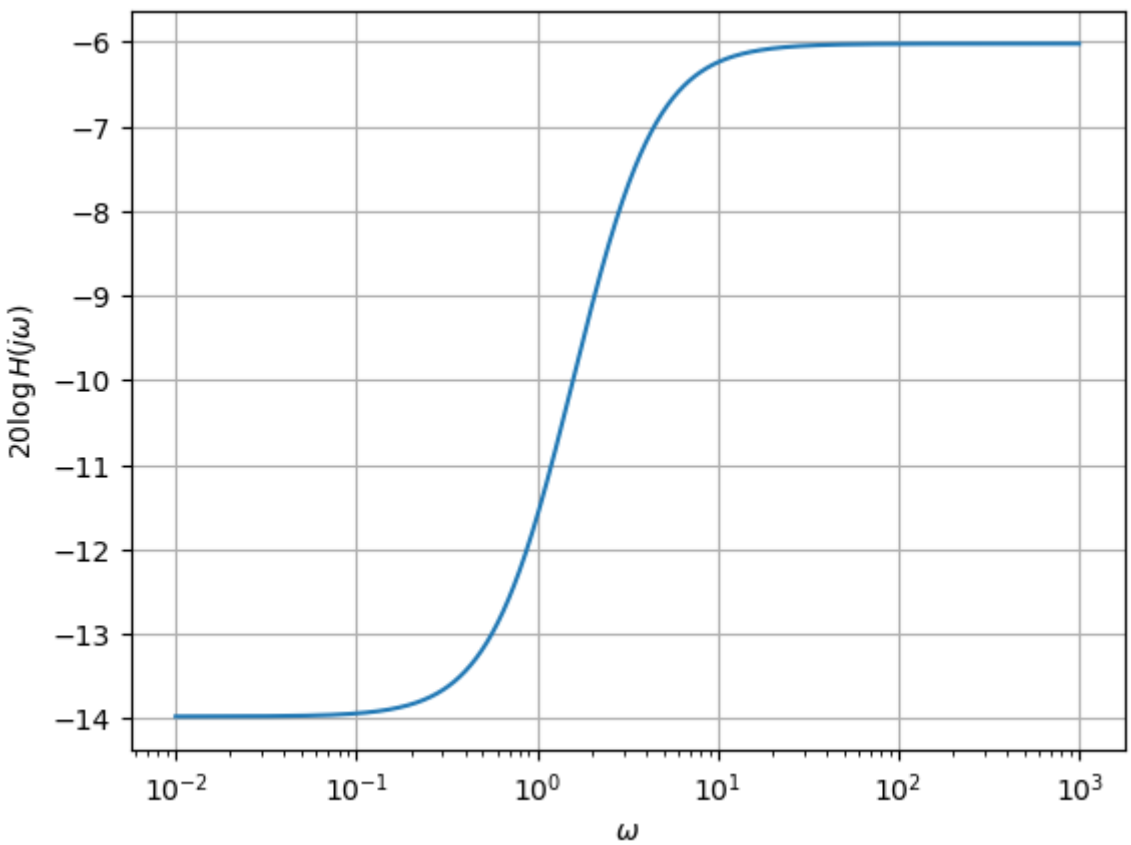
(ii)

The log magnitude is first found

$$\begin{aligned}
20\log(|H(j\omega)|) &= 20\log\left(\left|\frac{j\omega + 1}{2j\omega + 5}\right|\right) \\
&= 20\log\left(\frac{|j\omega + 1|}{|2j\omega + 5|}\right) \\
&= 20\log\left(\frac{\sqrt{1^2 + \omega^2}}{\sqrt{5^2 + (2\omega)^2}}\right) \\
&= 20\log\left(\frac{\sqrt{1 + \omega^2}}{\sqrt{25 + 4\omega^2}}\right)
\end{aligned}$$

The bode plot is then made in python using the following code:

```
omega = np.logspace(-2, 3, 1000)
H = 20 * np.log10(np.sqrt(1 + omega**2) / np.sqrt(25 + 4*omega**2))
plt.plot(omega, H)
plt.xlabel("$\omega$")
plt.ylabel("$20 \log H(j\omega)$")
plt.grid()
plt.semilogx()
```



(b)

(i)

Applying the Laplace transform to both sides using table 9.1 yields

$$2sY(s) + 5Y(s) = sX(s) + X(s)$$

The transfer function can then be found by isolating $\frac{Y(s)}{X(s)}$ resulting in

$$H(s) = \frac{s+1}{2s+5}$$

(ii)

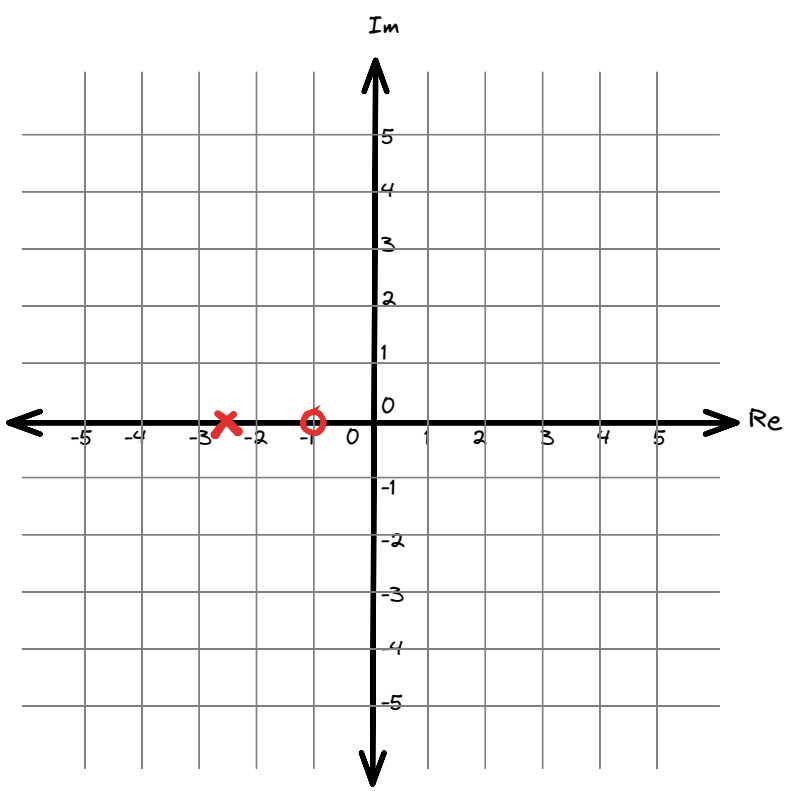
Zeros

The numerator will be zero if $s = -1$ since inserting this would yield $-1 + 1 = 0$.

Poles

The denominator will be zero if $s = -\frac{5}{2}$ since inserting this would yield $2 \cdot -\frac{5}{2} + 5 = -5 + 5 = 0$

Pole-zero plot



Question 5

(a)

The Nyquist rate is defined to be $2\omega_m$ where $\omega_m = 2\pi f_m$ is the highest angular frequency a given system works with. In this case the given system is a dogs hearing, which goes up to $f_m = 65\text{kHz}$ equating to a Nyquist rate of

$$\omega_s = 2 \cdot \omega_m = 2 \cdot 2\pi 65000 = 260000\pi$$

This makes the maximum sampling interval

$$T_s = \frac{2\pi}{\omega_s} = \frac{1}{130000}$$

(b)

(i)

For this signal, the frequency reached is $\omega_m = 4000\pi$, which makes the Nyquist rate $\omega_s = 8000\pi$ and the sampling period

$$T = T_s = \frac{1}{4000}$$

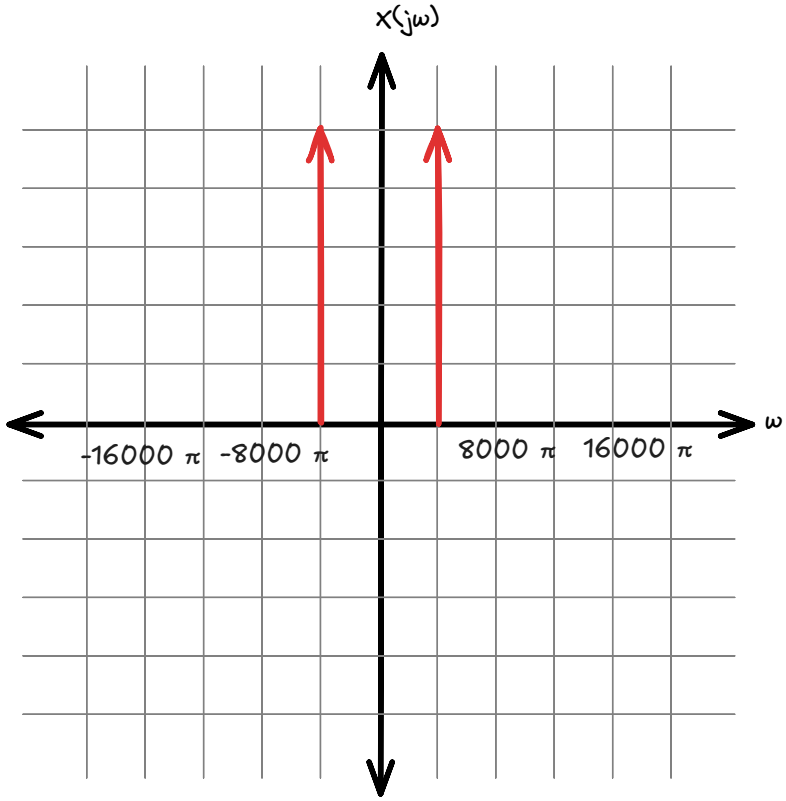
Now using table 4.2 we know

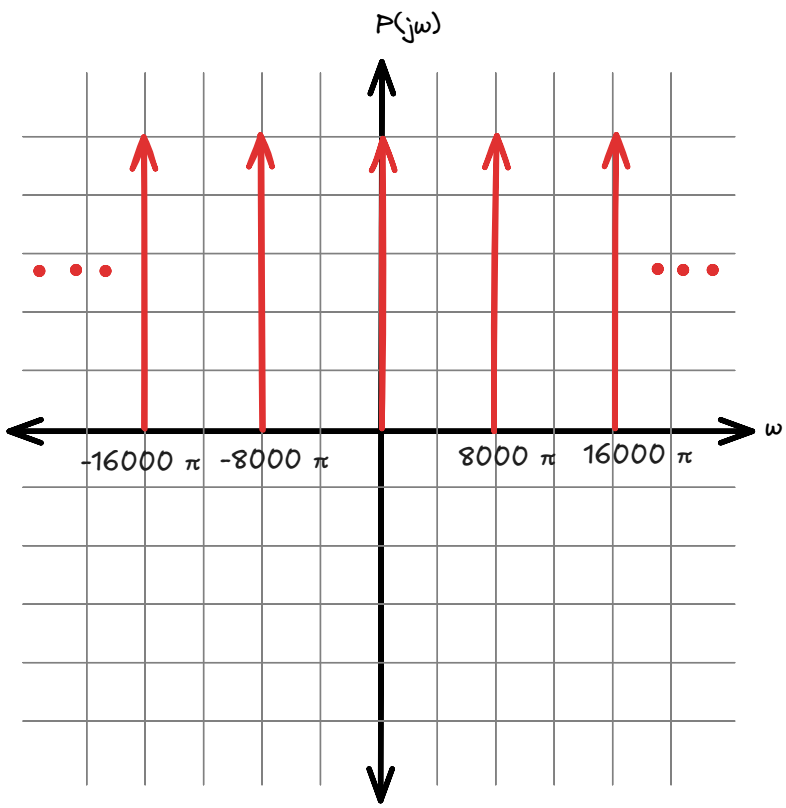
$$X(j\omega) = \pi(\delta(\omega - 4000\pi) - \delta(\omega + 4000\pi))$$

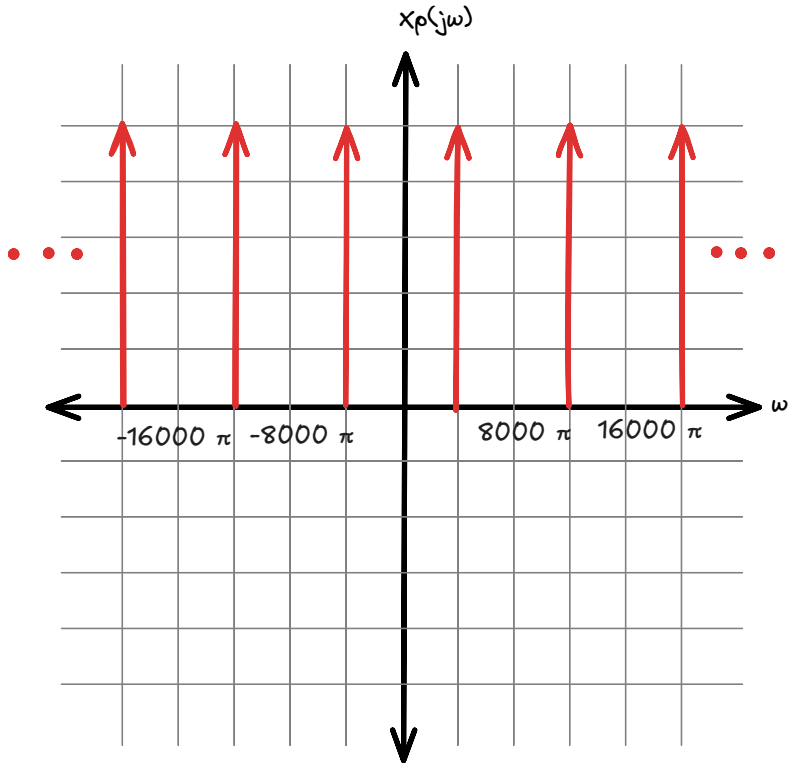
$$P(j\omega) = 8000\pi \cdot \sum_{k=-\infty}^{\infty} \delta(\omega - 8000\pi k)$$

$$\begin{aligned} X_p(j\omega) &= \frac{1}{2\pi}(X(j\omega) * P(j\omega)) \\ &= 4000 \sum_{k=-\infty}^{\infty} (\delta((\omega - 8000\pi k) - 4000\pi) - \delta((\omega - 8000\pi k) + 4000\pi)) \end{aligned}$$

The sketches of each signal would then be as follows


X(jω)
ω
-16000 π
-8000 π
8000 π
16000 π


P(jω)
ω
-16000 π
-8000 π
8000 π
16000 π



(ii)

As can be seen in the the sketch or mathematical equation of $X_p(j\omega)$, the Fourier transform does not contain any information in the interval $-3000\pi \leq \omega \leq 3000\pi$. This means that applying the low pass filter would lead to an outputted Fourier transform of

$$X_p(j\omega) = 0$$

This would give a corresponding sampled signal of

$$x_p(t) = 0$$

meaning that it does not contain any sinusoids.